

The two boundary values of the specified range, 0.01 and 0.1 to observe changes in the widest span of theta for an easier comparison. I used 1500 iterations for the entirety of this project.

Problem 1

1. Report the linear model you found for each explanatory variable.
 - a. $X1: 5.92794892 + -2.03833663 * X_1$
 - b. $X2: 0.73606043 + 0.55760761 * X_2$
 - c. $X3: 2.8714221 - 0.52048288 * X_3$
2. Plot the final regression model and loss over the iteration per each explanatory variable.
 - a. See pdf #2
3. Which explanatory variable has the lower loss (cost) for explaining the output (Y)?
 - a. X1 has the lowest cost for explaining the output (Y). This can be seen when comparing the cost history of X1 vs X2 and X3.

cost_history X1 = [5.16999006 4.96338989 4.7855721 ... 0.98499308
0.98499308 0.98499308]

cost_history X2 = [3.90731819 3.66528504 3.62832072 ... 3.59936602
3.59936602 3.59936602]

cost_history X3 = [4.66843939 4.49602325 4.43685075 ... 3.62945112
3.62945112 3.62945112]

We can see that the cost of X1 is just below one, while the cost for X2 and X3 are roughly the same at 3.6.

4. Based on your training observations, describe the impact of the different learning rates on the final loss and number of training iterations.
 - a. Starting with a modest 0.01 alpha (learning rate) we obtain the following:

Final value of theta X01 = [5.71850653 -1.9568206]

Final value of theta X02 = [0.71988473 0.56390334]

Final value of theta X03 = [2.78048129 -0.48451631]

cost_history X01 = [5.48226715 5.44290965 5.40604087 ... 0.99063932
0.99061433 0.99058944]

cost_history X02 = [5.29831663 5.09909109 4.92356115 ... 3.5993997
3.59939955 3.5993994]

cost_history X03 = [5.40768785 5.30397076 5.21178297 ... 3.63053597
3.6305311 3.63052625]

With a learning rate of 0.01 we can see that most of the values have converged, the values are still nearly identical nearing the end of iterations (1500) but X1 is still noticeably dropping. However, with a learning rate of $\alpha = 0.1$, we obtain the following:

Final value of theta X1 = [5.92794892 -2.03833663]

Final value of theta X2 = [0.73606043 0.55760761]

Final value of theta X3 = [2.8714221 -0.52048288]

cost_history X1 = [5.16999006 4.96338989 4.7855721 ... 0.98499308
0.98499308 0.98499308]

cost_history X2 = [3.90731819 3.66528504 3.62832072 ... 3.59936602
3.59936602 3.59936602]

cost_history X3 = [4.66843939 4.49602325 4.43685075 ... 3.62945112
3.62945112 3.62945112]

Here we can see that the cost history of X1 actually converged to 0.98499308, meaning that it took less iterations to reach a converged cost, compared to the $\alpha = .01$ run which had not yet fully converged by iteration 1500. This shows that the learning rate positively impacted the number of iterations needed to find the optimal theta value. Note that not every change to the learning rate is a good change, careful increments are required to prevent overshooting and instability. Overshoot was obtained when testing $\alpha = 0.8$, which caused the model to create a theta value "NaN"

Problem 2

1. Report the final linear model you found the best.
 - a. $5.31416563 - 2.00371905 \cdot X1 + 0.53256359 \cdot X2 - 0.26560164 \cdot X3$
2. Plot loss over the iteration.
 - a. See PDF #2
3. Based on your training observations, describe the impact of the different learning rates on the final loss and number of training iterations.
 - a. A modest learning rate of $\alpha = 0.01$ provided the following:

Final value of theta X_all_vars_01 = [4.15118728 -1.8394291 0.72473856
-0.09513266]

cost_history X_all_vars_01 = [5.21542243 4.97171977 4.7765543 ...
0.81079843 0.81065429 0.81051044]

A more aggressive learning rate of $\alpha = 0.1$ provided the following:

Final value of theta X_all_vars_1 = [5.31416563 -2.00371905 0.53256359
-0.26560164]

cost_history X_all_vars_1 = [4.13064348 3.51770697 3.12758306 ... 0.73846424
0.73846424 0.73846424]

Unlike in Problem 1 where both tested learning rates provided similar cost history values, this problem shows completely different cost history values for each α . When using an α of 0.01 we end with a cost value of 0.8105, which had not yet converged by iteration 1500. Using $\alpha = 0.1$, the cost converged to 0.738, reaching a lower stable value within the same 1500 iterations. This suggests $\alpha = .01$ would require more than 1500 iterations to reach the same converged cost that $\alpha = 0.1$ obtained

4. Predict the value of y for new (X_1, X_2, X_3) values (1, 1, 1), for (2, 0, 4), and for (3, 2, 1)
 - a. 3.57740853
 - b. 0.24432097
 - c. 0.10253402